

2020 HKDSE Physics

Paper 1A — Multiple-choice Questions

Editable transcription of all questions and formulae. Only necessary, non-duplicated original diagrams are inserted. Each official answer is followed by a verified concise full-mark solution in English, with editable subscripts and superscripts.

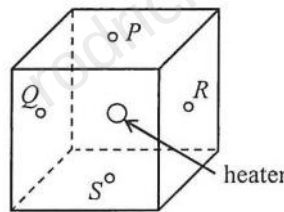
Editable transcription with official answers and verified concise full-mark reasoning in English.

Questions

Official answers are placed directly below each question.

Question 1

A heater is installed at the centre of a fully filled cubic water tank. Temperature sensors P, Q, R and S are fixed at the respective centres of the top, left, right and bottom surfaces of the tank. After the heater is switched on for a short duration, which pair of sensors below would indicate the largest temperature difference?



- A. Q and R
- B. R and S
- C. Q and S
- D. P and R

Official answer: D

The heater is at the centre. Convection carries hot water upward to P, while the cooler return flow is near R, so the largest temperature difference is between P and R. Answer: D.

Question 2

An electric kettle which contains 1 kg of water at room temperature takes 168 s to heat up the water to boiling point. The kettle's rated value is '220 V, 2000 W'. Assume that all the electrical energy consumed by the kettle is transferred to the water. Which of the following statements is/are correct? Given: specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$ (1) The initial temperature of the water is 20°C . (2) The resistance of the kettle's heating element is about 24Ω . (3) If the electric kettle is operated with 110 V, the time taken to heat up the water to boiling point will be doubled.

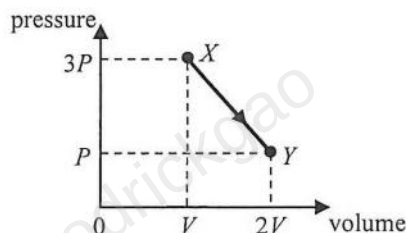
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (1), (2) and (3)

Official answer: C

$E = Pt = 2000(168) = 336 \text{ kJ} = mc(100 - T_i)$, giving $T_i = 20^{\circ}\text{C}$. Also $R = V^2/P = 220^2/2000 \approx 24 \Omega$. At 110 V the same heater has one-quarter of its original power, so the heating time would be four times, not twice, as long. Thus (1) and (2) are correct. Answer: C.

Question 3

A fixed mass of an ideal gas expands from state X to state Y through a process as represented in the pressure–volume graph below. If the temperature of the gas at state Y is 25°C , what is its temperature at state X?



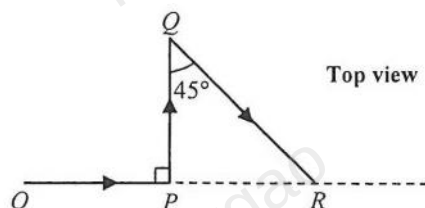
- A. -74.3°C
- B. 16.7°C
- C. 37.5°C
- D. 174°C

Official answer: D

For a fixed amount of gas, T is proportional to pV . From the graph, X has pressure $3p$ and volume V , while Y has pressure p and volume $2V$. Hence $T_X/T_Y = 3/2$. With $T_Y = 298 \text{ K}$, $T_X = 447 \text{ K} = 174^{\circ}\text{C}$. Answer: D.

Question 4

A car takes 8 minutes to travel along a path OPQR on a horizontal surface as shown. Given that $OP = PQ = 2$ km, find the magnitude of the average velocity of the car in this journey.



- A. 30 km h^{-1}
- B. 36 km h^{-1}
- C. 41 km h^{-1}
- D. 51 km h^{-1}

Official answer: A

OP is 2 km east and PQ is 2 km north. QR is $2\sqrt{2}$ km at 45° downward, so it contributes 2 km east and 2 km south. The resultant displacement is therefore 4 km east. Since $8 \text{ min} = 2/15 \text{ h}$, the average velocity is $4/(2/15) = 30 \text{ km h}^{-1}$ east. Answer: A.

Question 5

For a car travelling on a highway, which of the following statements about the safety design of the headrest is/are correct? (1) As the headrest is soft, it can reduce the force exerted on the passenger's head during impact. (2) It can minimise injury to the passenger when the car is struck from behind. (3) It can minimise injury to the passenger when the car brakes suddenly.



- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: C

A soft headrest increases the stopping time of the head and reduces the force. In a rear collision it also prevents excessive backward motion of the head relative to the body. During sudden braking the head moves forward, so the headrest behind it does not provide the claimed protection. Thus (1) and (2) are correct. Answer: C.

Question 6

A particle is thrown vertically upward and its path is shown below. H is the highest point that the particle reached. Which of the following statements about the particle is/are correct? Neglect air resistance. (1) Its acceleration at M is upward. (2) Its acceleration at H is zero. (3) Its acceleration at N is downward.



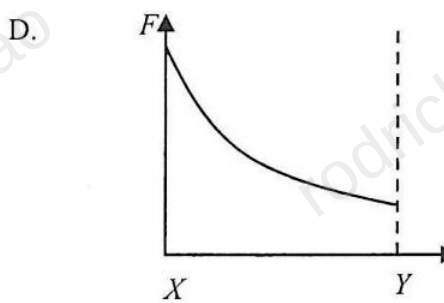
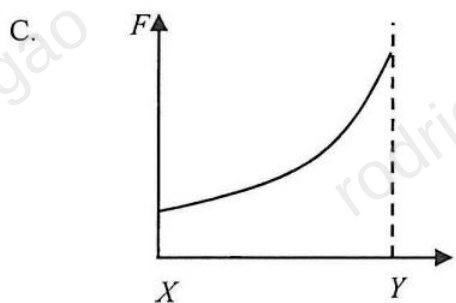
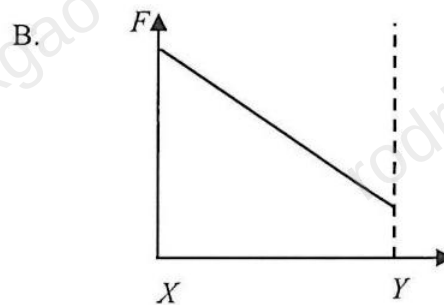
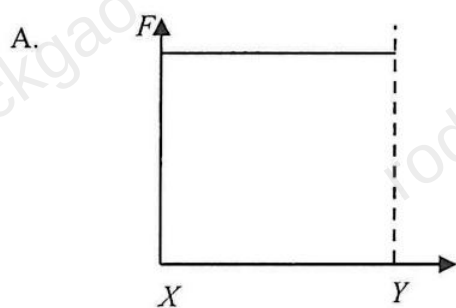
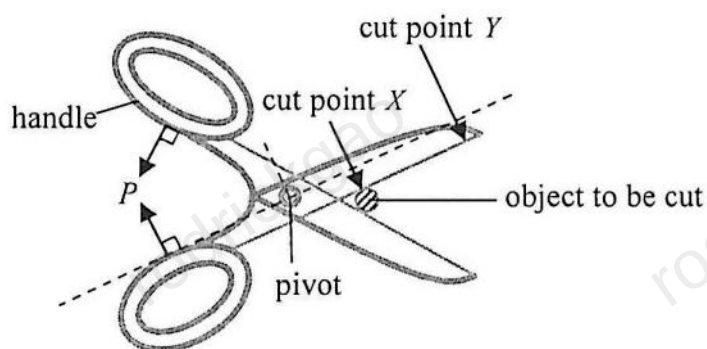
- A. (1) only
- B. (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: B

At M, H and N the acceleration due to gravity is always vertically downward, including at the highest point H where the vertical velocity is momentarily zero. Only statement (3) is correct. Answer: B.

Question 7

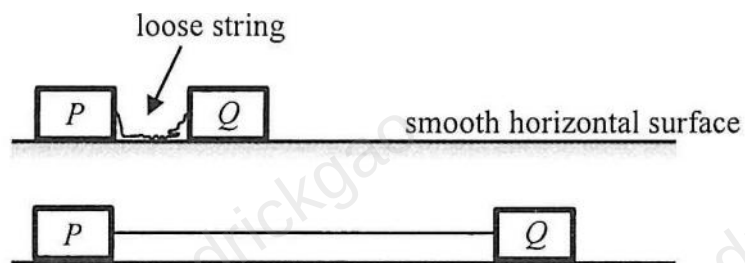
The figure shows that a pair of forces P of constant magnitude is applied at right angles to the pair of scissors' handles in order to cut an object. Which graph below best shows the variation of force F produced at the cut point from X to Y when the pair of scissors is closed?

**Official answer: D**

The applied force P is perpendicular to the handle and acts at a fixed distance from the pivot, so its turning moment is constant. The cutting force is inversely proportional to the perpendicular distance of the cutting point from the pivot; it therefore decreases non-linearly from X to Y . Answer: D.

Question 8

On a smooth horizontal surface, two identical blocks P and Q are connected by a light inextensible string. The blocks are at rest and the string is loose initially. Q is given a speed of 4 m s^{-1} and moves to the right. Find the speeds of the blocks when the string just becomes taut and P starts to move.



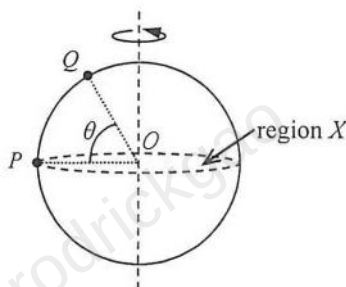
- A. P: 1 m s^{-1} ; Q: 1 m s^{-1}
- B. P: 2 m s^{-1} ; Q: 1 m s^{-1}
- C. P: 2 m s^{-1} ; Q: 2 m s^{-1}
- D. P: 4 m s^{-1} ; Q: 2 m s^{-1}

Official answer: C

When the string becomes taut, the impulsive tension makes the two equal masses move together instantaneously. Conservation of momentum gives $m(4) = 2mv$, so $v = 2 \text{ m s}^{-1}$ for both masses. Answer: C.

Question 9

Particles P and Q are situated on a rotating spherical body as shown. The body rotates uniformly about the axis XY. Region X is a hemisphere and its edge is a circle with P situated at the edge. The angle of elevation of Q from the centre O is θ . Find the ratio of the centripetal acceleration of P to that of Q.

A. 1: $\cos \theta$ B. 1: $\sin \theta$ C. $\cos \theta$: 1D. $\sin \theta$: 1**Official answer: A**

For rotation about the axis, centripetal acceleration is $a = \omega^2 r_{\perp}$. P is at perpendicular distance R, whereas Q is at $R \cos \theta$. Therefore $a_P : a_Q = 1 : \cos \theta$. Answer: A.

Question 10

The diameter and mass of Neptune are respectively about 4 times and 17 times those of the Earth. Estimate the acceleration due to gravity on the surface of Neptune. Given: acceleration due to gravity on the surface of the Earth = 9.81 m s^{-2} .

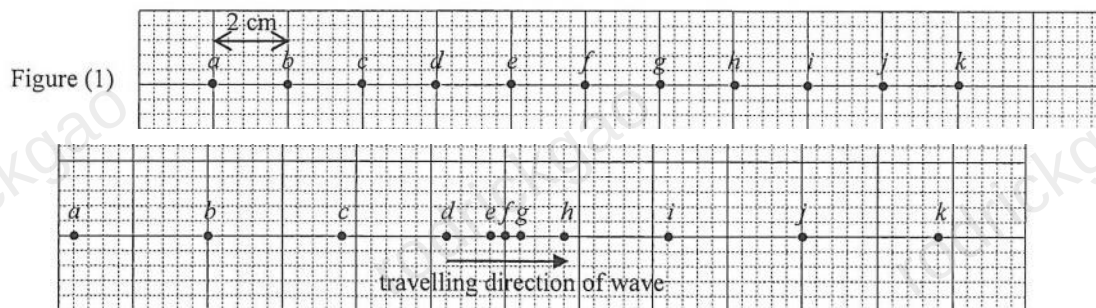
A. 2.3 m s^{-2} B. 9.2 m s^{-2} C. 10.4 m s^{-2} D. 41.7 m s^{-2} **Official answer: C**

Using g proportional to M/R^2 , $g_N/g_E = 17/4^2 = 17/16$. Hence $g_N \approx (17/16)(9.81) = 10.4 \text{ m s}^{-2}$. Answer: C.

Question 11

Figure 1 shows the equilibrium positions of particles *a* to *k* in a medium. Adjacent particles are separated by 2 cm. A longitudinal wave of frequency 5 Hz travels from left to right in the medium. Figure 2 shows the positions of the particles at a certain instant. Determine the amplitude and the speed of the wave.

11.



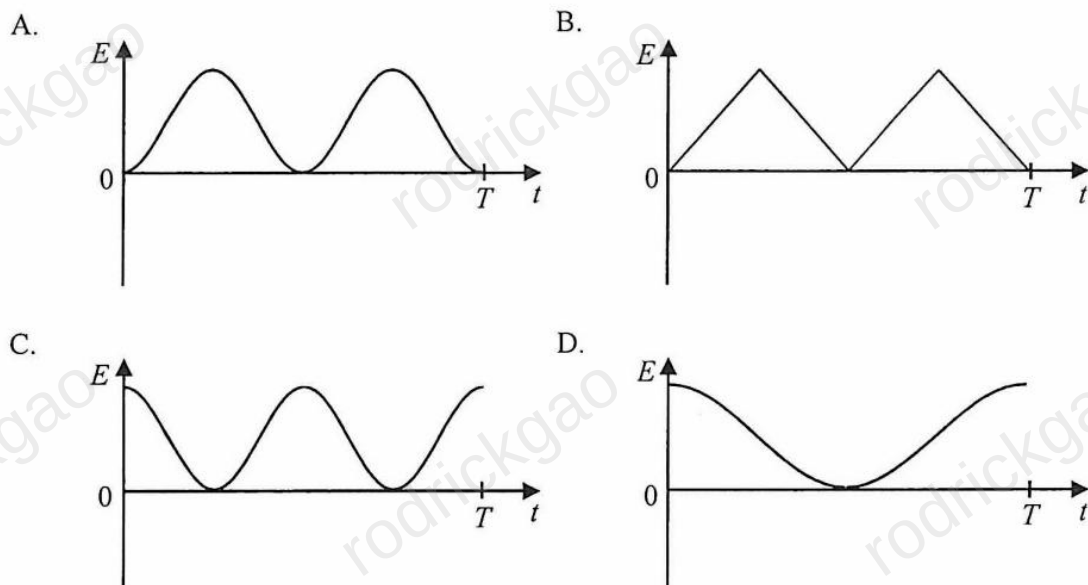
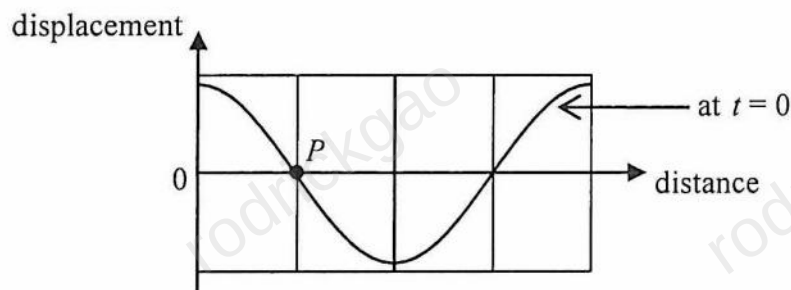
- A. Amplitude: 3.6 cm; speed: 40 cm s⁻¹
- B. Amplitude: 3.6 cm; speed: 80 cm s⁻¹
- C. Amplitude: 2.4 cm; speed: 40 cm s⁻¹
- D. Amplitude: 2.4 cm; speed: 80 cm s⁻¹

Official answer: D

The graph gives an amplitude of 2.4 cm and a wavelength of 16 cm. Thus $v = f\lambda = 5(16) = 80 \text{ cm s}^{-1}$. Answer: D.

Question 12

The figure shows part of the displacement–distance graph of a travelling wave of period T at $t = 0$. P is a particle on the wave. Which graph below correctly shows the variation of the particle's kinetic energy E within a period starting from $t = 0$?

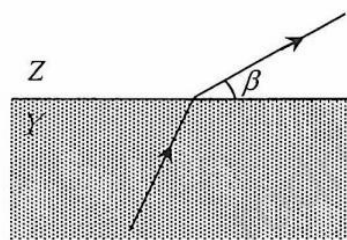
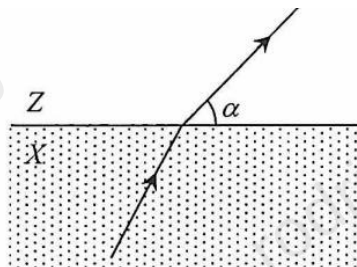


Official answer: C

At P the displacement is zero, so the oscillator has maximum speed and maximum kinetic energy. In SHM the kinetic energy repeats every $T/2$, beginning at a maximum and reaching two minima in one period. Answer: C.

Question 13

Monochromatic light travels with the same incident angle from media X and Y respectively to another medium Z as shown. The corresponding refracted rays in Z make angles α and β respectively with the boundary plane, where $\alpha > \beta$. Which medium, X or Y, has a greater refractive index? In which medium, X or Y, does light travel faster?



- A. Greater refractive index: X; light travels faster in: X
- B. Greater refractive index: X; light travels faster in: Y
- C. Greater refractive index: Y; light travels faster in: X
- D. Greater refractive index: Y; light travels faster in: Y

Official answer: C

The incidence angles are equal. Snell's law gives $n_X \sin i = n_Z \cos \alpha$ and $n_Y \sin i = n_Z \cos \beta$. Since $\alpha > \beta$, $\cos \alpha < \cos \beta$ and hence $n_Y > n_X$; light therefore travels faster in X. Answer: C.

Question 14

Monochromatic light of wavelengths λ_1 and λ_2 is incident normally in turn on the same diffraction grating. The third-order maximum produced by λ_1 coincides with the fourth-order maximum produced by λ_2 . If $\lambda_1 = 680$ nm, find λ_2 .

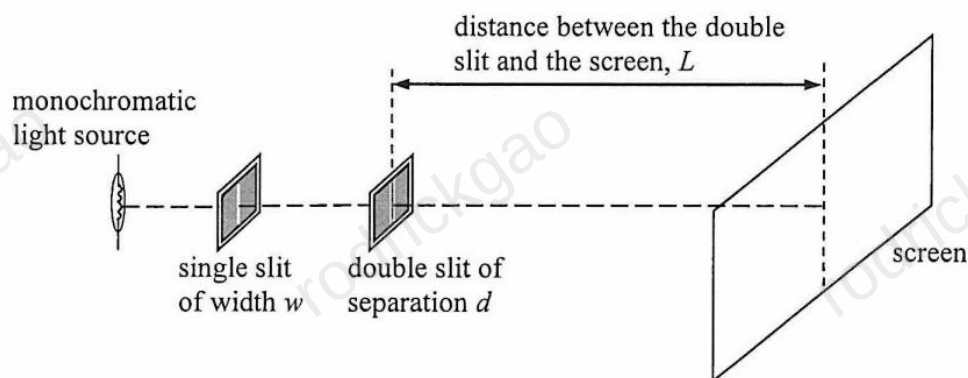
- A. 510 nm
- B. 680 nm
- C. 907 nm
- D. Cannot be determined

Official answer: A

Coincident grating maxima satisfy $m_1 \lambda_1 = m_2 \lambda_2$. From the shown orders, $3\lambda_1 = 4\lambda_2$. Hence $\lambda_2 = 3(680)/4 = 510$ nm. Answer: A.

Question 15

The figure shows a typical set-up of Young's double-slit experiment. Which combination below is the best setting for displaying an observable fringe pattern on the screen?



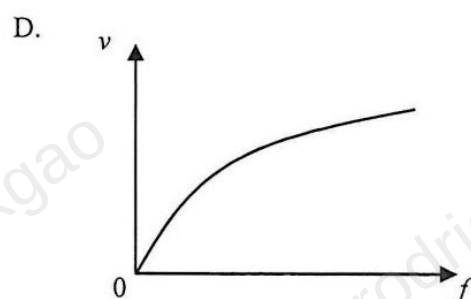
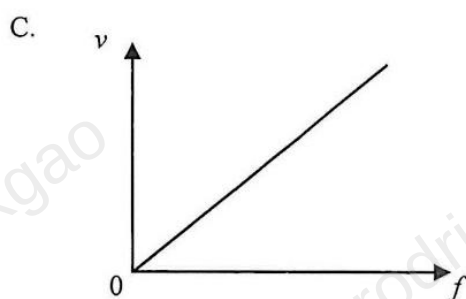
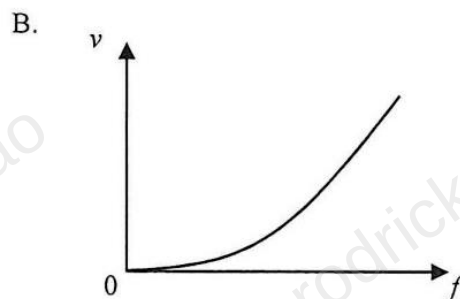
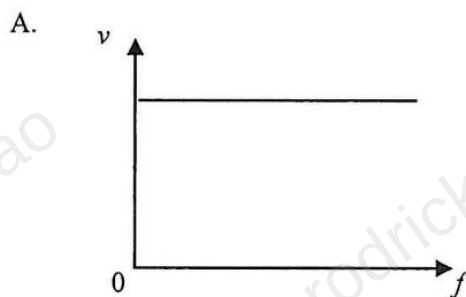
- A. $w = 0.1 \text{ mm}$; $d = 1 \text{ mm}$; $L = 10 \text{ m}$
- B. $w = 0.1 \text{ mm}$; $d = 1 \text{ mm}$; $L = 1 \text{ m}$
- C. $w = 1 \text{ mm}$; $d = 0.1 \text{ mm}$; $L = 1 \text{ m}$
- D. $w = 1 \text{ mm}$; $d = 0.1 \text{ mm}$; $L = 0.1 \text{ m}$

Official answer: C

Fringe spacing is $\beta = \lambda L/d$. Arrangement C gives sufficiently separated fringes at a practical screen distance, while the wider single slit transmits more light so that the pattern is easier to observe. Answer: C.

Question 16

A transverse wave propagates along a stretched string. Which graph below correctly shows the variation of the speed v of the wave with its frequency f ?

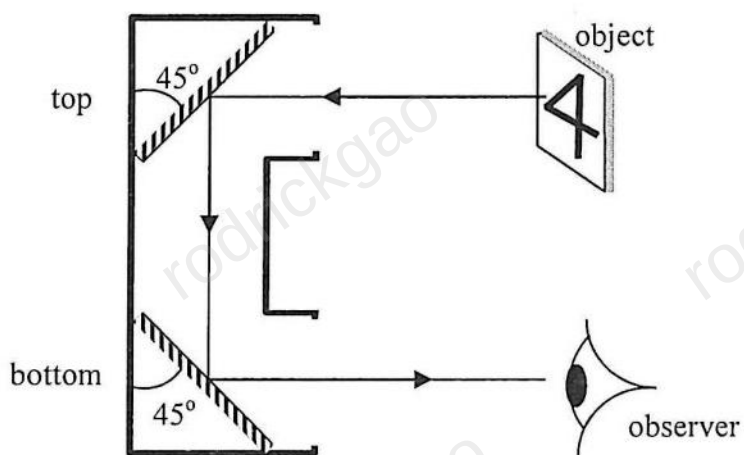


Official answer: A

For a stretched string, $v = \sqrt{T/\mu}$. With the same string and tension, v is independent of driving frequency, so the speed–frequency graph is horizontal. Answer: A.

Question 17

The figure shows a periscope designed by a student. An object is observed via the periscope. Which image will the observer see?



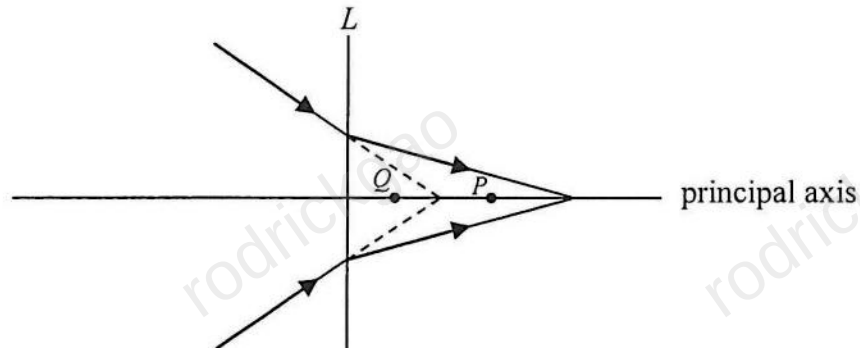
- A.
- B.
- C.
- D.

Official answer: D

The two reflecting planes are perpendicular. Two successive reflections in perpendicular mirrors rotate the image through 180° , giving orientation D. Answer: D.

Question 18

Referring to the ray diagram, what kind of lens is represented by L? Which point, P or Q, is a focus of L?



- A. Concave lens; P
- B. Convex lens; P
- C. Concave lens; Q
- D. Convex lens; Q

Official answer: A

The emerging rays are less convergent than the incident rays, so L is a diverging concave lens. Extending the rays according to the construction locates P as the focal point. Answer: A.

Question 19

Which of the following phenomena would provide conclusive evidence for the wave nature of sound? (1) reflection (2) refraction (3) interference

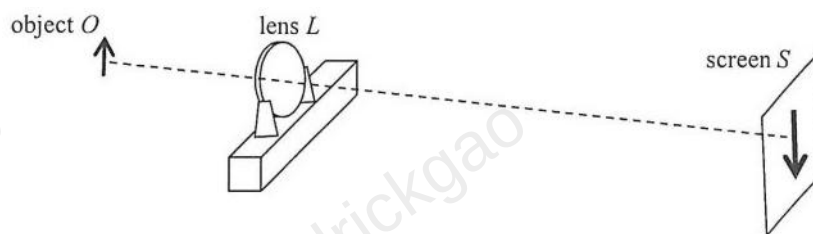
- A. (2) only
- B. (3) only
- C. (1) and (2) only
- D. (1) and (3) only

Official answer: B

Interference is direct evidence of wave behaviour. Reflection and refraction can also be described using particle models, so only statement (3) is decisive. Answer: B.

Question 20

An enlarged sharp image of an object O is formed on a screen S by a convex lens L as shown. Which of the following methods can produce a diminished sharp image on the screen? (1) Fix O and L, and move S closer to L. (2) Fix L and S, and move O farther from L. (3) Fix O and S, and move L closer to S.



- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: B

The original real image is enlarged, so $f < u < 2f$. With object and screen fixed, moving the lens toward the screen reaches the conjugate position and forms a diminished real image. Only statement (3) is correct. Answer: B.

Question 21

Which of the following statements about ultrasound is/are correct? (1) It has a shorter wavelength than audible sound in the same medium. (2) It cannot be produced by vibrating objects. (3) It cannot be heard by humans because it cannot travel through air.

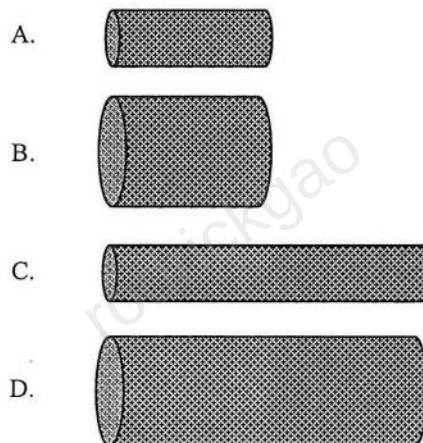
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: A

Ultrasound has a higher frequency than audible sound. In the same medium its speed is essentially unchanged, so $\lambda = v/f$ is shorter. Ultrasound can be produced by a vibrating source and can travel through air. Only statement (1) distinguishes it. Answer: A.

Question 22

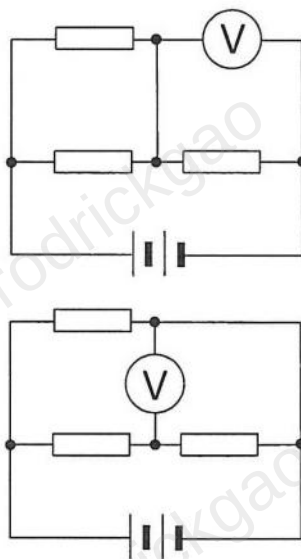
The cylindrical resistors shown are made of the same material. Which resistor dissipates the greatest power when the same voltage is applied across it?

**Official answer: B**

For a wire, $P = V^2/R$ and $R = \rho L/A$. Wire B is the shortest and thickest, so it has the smallest resistance and therefore the greatest power at the same voltage. Answer: B.

Question 23

Three identical resistors, a battery of negligible internal resistance and an ideal voltmeter are connected to form Circuits (a) and (b) respectively. Given that the voltmeter reading is 8 V in Circuit (a), what is the voltmeter reading in Circuit (b)?



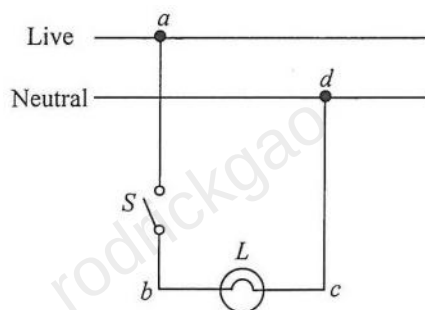
- A. 4 V
- B. 6 V
- C. 8 V
- D. 12 V

Official answer: B

In the first circuit, the two parallel resistors have resistance $R/2$ and are in series with R . If the series resistor has 8 V across it, the parallel part has 4 V, so the supply is 12 V. In the second circuit, two equal series resistors share 12 V equally, and the voltmeter reads 6 V. Answer: B.

Question 24

The figure shows part of a domestic lighting circuit. Bulb L does not light up when switch S is closed. A voltage tester gives high readings at b and c. At a and d, the tester gives a high reading only at a. What is the most likely fault?



- A. Switch S is damaged.
- B. The filament of bulb L is burnt and forms an open circuit.
- C. There is a short circuit between a and d.
- D. There is an open circuit between c and d.

Official answer: D

Opening c–d leaves point c connected to the live wire through the intact lamp. Thus a, b and c remain at high potential, while d is connected to neutral. The matching set is D. Answer: D.

Question 25

A fully charged battery is labelled '3.7 V, 1100 mA h'. It supplies a current of 250 mA for one hour. Which expression gives the energy supplied by the battery during this hour?



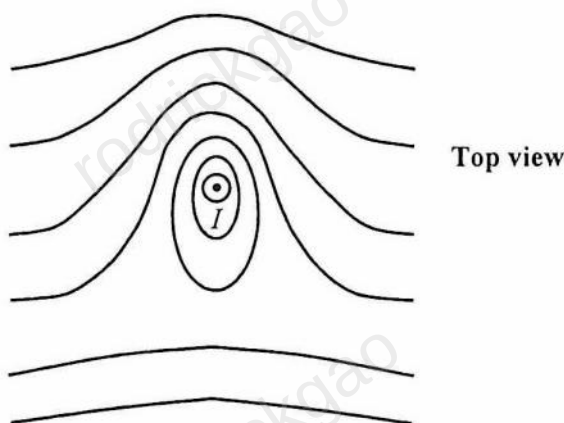
- A. $(3.7 \times 250/1000 \times 3600) \text{ J}$
- B. $(3.7 \times 1100/1000 \times 3600) \text{ J}$
- C. $(3.7 \times 250/1000 \times 1) \text{ J}$
- D. $(3.7 \times 1100/1000 \times 1) \text{ J}$

Official answer: A

The energy delivered is $E = VIt = 3.7(0.250)(3600) = 3.33 \times 10^3 \text{ J}$. The 250 mA h used is also below the 1100 mA h battery capacity. Answer: A.

Question 26

The figure shows the resultant magnetic field pattern around a long vertical current-carrying wire. The current flows out of the paper. The Earth's magnetic field cannot be neglected. Which option correctly shows the direction of the horizontal component of the Earth's magnetic field and the direction of the magnetic force acting on the wire?



the horizontal component of
the Earth's magnetic field

the magnetic force experienced
by the current-carrying wire

- A.
B.
C.
D.

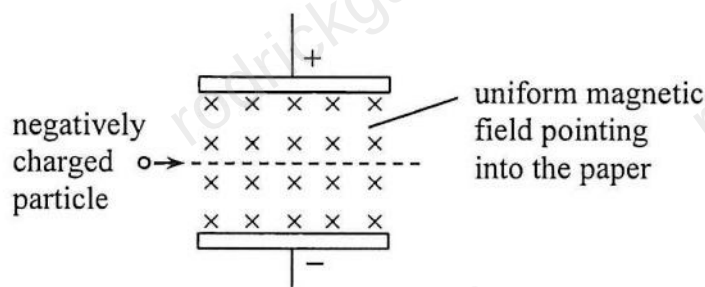


Official answer: A

The compass pattern shows the Earth's horizontal magnetic field directed to the left. With current out of the page, $F = IL \times B$ gives a downward magnetic force. Answer: A.

Question 27

A negatively charged particle passes undeflected through mutually perpendicular electric and magnetic fields as shown. Which of the following changes would cause the particle to deflect downwards? (1) Increase the potential difference across the plates. (2) Increase the magnitude of the charge on the particle. (3) Increase the speed of the particle.



- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

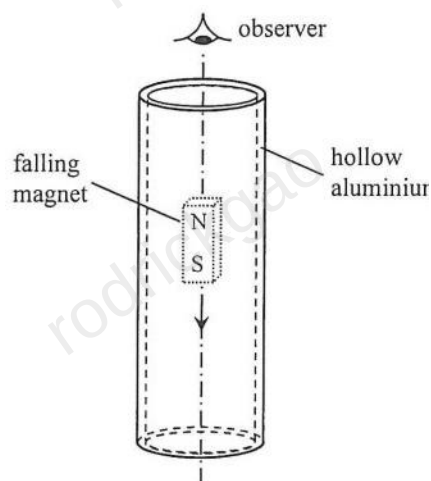
Official answer: B

Initially the electric and magnetic forces balance. Increasing the speed increases qvB but not qE , so the particle is deflected downward. Changing the charge magnitude scales both forces equally and preserves the balance.

Only statement (3) is correct. Answer: B.

Question 28

A bar magnet falls vertically through a hollow aluminium tube. Viewed from above, what are the directions of the induced current in the tube above and below the magnet?



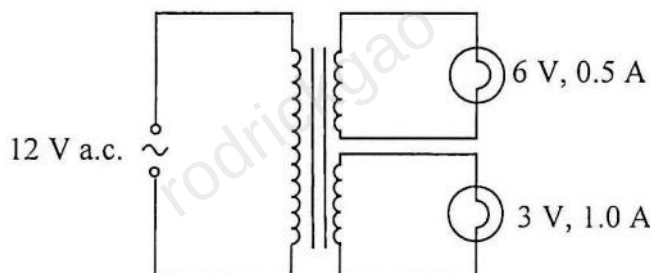
- A. Clockwise above; clockwise below
- B. Anticlockwise above; anticlockwise below
- C. Clockwise above; anticlockwise below
- D. Anticlockwise above; clockwise below

Official answer: D

As the north pole approaches from above, the downward magnetic flux through the ring increases, so an anticlockwise current produces an opposing upward field. After the magnet passes, the downward flux from the departing south pole decreases, so a clockwise current produces a downward field to maintain it. Answer: D.

Question 29

An ideal transformer has two secondary coils supplying two bulbs rated 6 V, 0.5 A and 3 V, 1 A respectively. The primary is connected to a 12 V a.c. supply. What is the current in the primary coil?



- A. 0.25 A
- B. 0.50 A
- C. 0.75 A
- D. 1.0 A

Official answer: B

The total secondary output power is $6(0.5) + 3(1.0) = 6.0$ W. For an ideal transformer, $12I_p = 6.0$, giving $I_p = 0.50$ A. Answer: B.

Question 30

A Geiger–Müller counter is used to measure the background count rate. Which set of four readings is most reasonable?

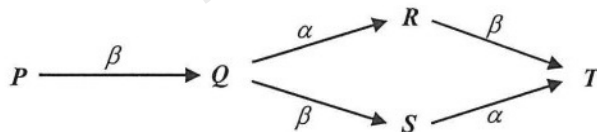
- A. 5, 62, 8, 69
- B. 40, 40, 40, 40
- C. 60, 50, 30, 20
- D. 29, 26, 31, 35

Official answer: D

Background count is random and fluctuates about a steady mean according to counting statistics. The sequence 29, 26, 31 and 35 shows a plausible random spread without a systematic trend. Answer: D.

Question 31

The figure shows a radioactive decay chain. Which deductions are correct? (1) P and T are isotopes. (2) Q and S have the same number of protons. (3) S has one more neutron than R.



- A. (1) and (2) only
- B. (1) and (3) only**
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: B

From P to T, the proton number is unchanged while the nucleon number falls by 4, so P and T are isotopes. S is formed from Q by β decay and therefore has a different proton number. Counting A–Z shows that S has one more neutron than R. Thus (1) and (3) are correct. Answer: B.

Question 32

Which statement about the decay constant of a radioactive isotope is correct?

- A. It varies randomly with time.
- B. It depends on pressure and temperature.
- C. It is directly proportional to the number of nucleons in the nucleus.
- D. It is a characteristic quantity that can be used to identify the isotope.**

Official answer: D

The decay constant is an intrinsic property of a radionuclide. It is independent of temperature, pressure and the amount of sample, so it is the identifying quantity listed. Answer: D.

Question 33

After the same period of time, $15/16$ of the nuclei in a sample of radioactive isotope P have decayed, while $63/64$ of the nuclei in a sample of radioactive isotope Q have decayed. Find the ratio of the half-life of P to that of Q.

- A. 1: 4
- B. 2: 3
- C. 3: 2
- D. 4: 1

Official answer: C

For P, the remaining fraction $1/16 = (1/2)^4$, so $t = 4T_P$. For Q, $1/64 = (1/2)^6$, so $t = 6T_Q$. Hence $T_P/T_Q = 6/4 = 3/2$.
Answer: C.